

Exploratory Data Analysis (EDA) for Data Science Projects

Filled template for the *meta-analysis-of-single-case-psy-studies* repository. This report summarizes the available study metadata, the linked paper corpus, and the main distributions that matter for downstream extraction and analysis.

Corpus

Metric	Value
Usable metadata rows	316
Source metadata variables	24
Derived analysis variables	2
Unique studies	71
PDF files in input/	71
Extracted text block JSON files	139
Distinct intervention labels in EDA output	35
Distinct demographic labels in EDA output	46

- The main structured source is **Supplement 3. Case and study characteristics.xlsx**. After removing empty rows and columns, the working table contains **316** rows, **24** original variables, and 2 derived analysis variables.
- The repository contains **71** source papers in **input/**. The matching logic in the repo links all study entries to a corresponding PDF, so PDF coverage is complete.
- There are **139** text-block extraction files in **extracted_text/**, which indicates that some papers were processed under more than one filename variant.

Data Cleaning And Provenance

- Empty rows and fully empty columns were removed before analysis.
- The **Study** column was forward-filled because a single study can span multiple case rows in the Excel sheet.
- Age was coerced to numeric to make range and central tendency estimates possible.
- Detailed anxiety diagnoses were also mapped to a broader **Anxiety Disorder** category for higher-level summaries.
- Outputs are preserved in **eda_output/**, including plots, frequency tables, and missing-value reports.

Missingness And Variable Quality

- **Treatment protocol**: 83.5% missing
- **Type of treatments(s)**: 77.8% missing
- **Setting University clinic, Outpatient Psychiatric Center, School, Other, Not reported**: 77.5% missing
- **Type of publication**: 77.5% missing
- **Quality rating RoBiNT scale Total, Internal validity, External validity**: 77.5% missing
- **Type of SCED design**: 77.5% missing
- **Sample type Referred, Recruited, Combination, Other, Not reported**: 77.5% missing
- **Drop-outs** : 77.5% missing

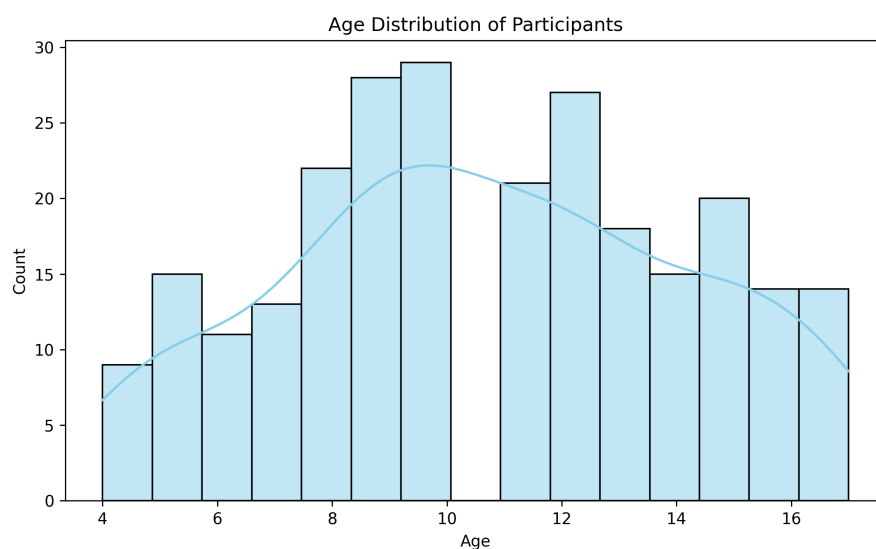
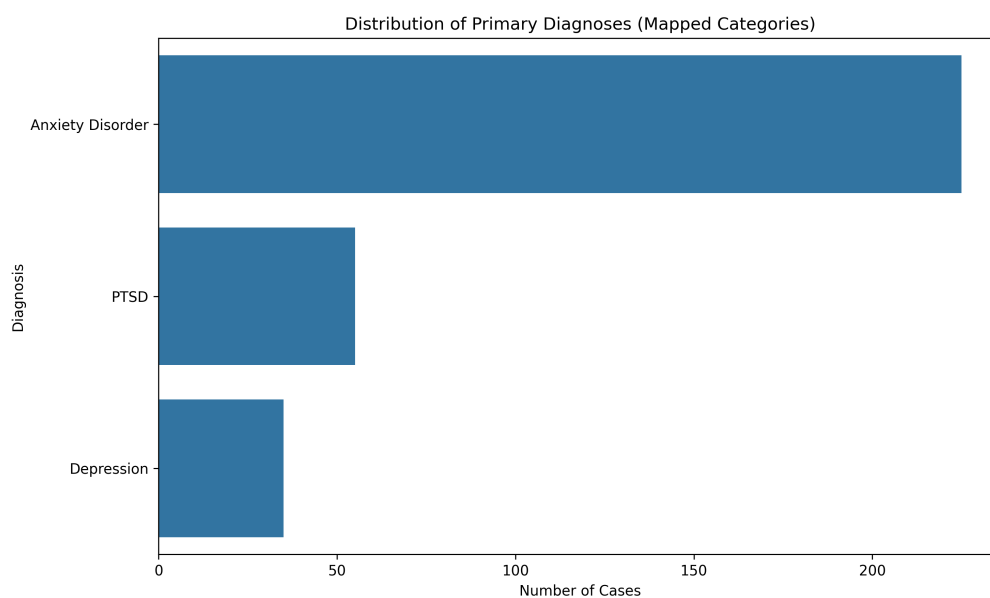
Missingness is concentrated in study-level descriptors such as treatment protocol, intervention type, country, setting, publication type, and SCED design. This matters because it limits how far multivariate comparisons can be pushed without first standardizing and completing metadata.

Pre Variable Description

- Participant ages are available for **256** of 316 rows (81.0%), with mean **10.72**, median **11.0**, and range **4-17** years.
- Explicit single-gender labels show **147** female and **120** male cases, with **39** missing and several mixed encodings such as 8F/2M or 4F/4M.
- Intervention and demographic frequencies were exported to CSV so the label space can be normalized before modeling.

Univariate Analysis

- The broad diagnosis distribution is dominated by **Anxiety Disorder** with **225** rows (71.2%).
- The next largest groups are **PTSD** (55 rows) and **Depression** (35 rows).
- The intervention frequency table is led by CBT and behavioral treatment variants, which suggests the corpus is methodologically concentrated rather than evenly spread across many therapy families.
- Among non-empty design entries, multiple baseline designs appear most often, but the design field itself is missing for most rows.



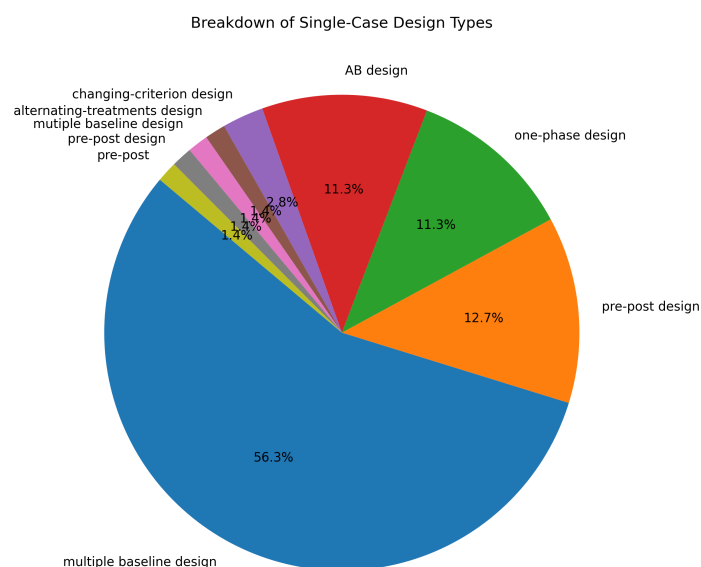
Baseline

If a model were asked to predict the broad diagnosis category using a naive majority-class rule, it could always predict **Anxiety Disorder** and achieve a baseline accuracy of **71.2%**. Any downstream classifier should beat that benchmark to be meaningful.

Multivariate Analysis

- AD (specific phobia): mean age **9.2** across **44** observed cases.
- PTSD: mean age **10.3** across **43** observed cases.
- AD (separation anx): mean age **8.4** across **37** observed cases.
- MDD: mean age **12.2** across **31** observed cases.
- AD (social anx): mean age **12.9** across **27** observed cases.
- AD (generalized anx): mean age **14.0** across **24** observed cases.

The age summaries suggest clinically different age profiles across diagnoses: generalized anxiety and social anxiety skew older, while separation anxiety and selective mutism skew younger. A full correlation analysis would require further label cleaning because categorical values currently use many inconsistent spellings.



Normalization Recommendations

- Collapse synonymous treatment labels such as CBT, Cognitive behavioral treatment, and CBT principles.
- Standardize SCED design spellings such as multiple baseline design, mutiple baseline design, and pre-post variants.
- Split combined gender strings into separate count variables so group totals are machine-readable.
- Normalize ethnicity spelling and casing because the current labels mix white, caucasian, New Zealand European, and many one-off encodings.
- Deduplicate extraction artifacts where the same paper produced more than one block JSON file because of filename variation.

Conclusion

This repository already supports a strong descriptive EDA workflow: the metadata table is linked to the paper corpus, the main demographic and intervention summaries are exported, and the plots make the dominant diagnosis and age patterns easy to inspect. The main limitation is metadata completeness and label consistency.